

Router

2: We should enable the telnet protocol

→ Router (config) # line vty 0 4
↓

5 devices can take

→ Router (config-line) # the router remotely
transport input telnet

→ " # login local

" # exit

3: Go to PC → Putty application

→ telnet → Enter Router IP.

Day-9: ROUTE TABLE

Like router, in PC also if we do IP configuration automatically values are stored.

→ To check this, → CMD → "Route print"
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→ This table is called system route table

SYSTEM

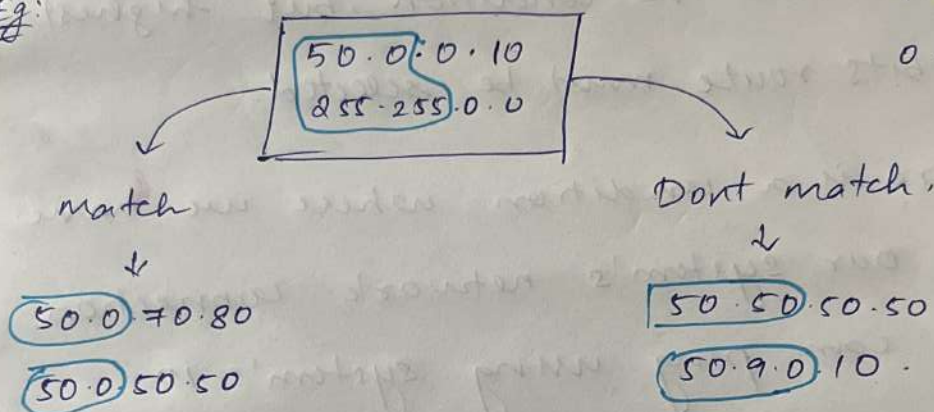
50.0.0.10

### ROUTE TABLE

| Network   | Netmask         | Gateway   |
|-----------|-----------------|-----------|
| 50.0.0.0  | 255.0.0.0       | 50.0.0.10 |
| 50.0.0.10 | 255.255.255.255 | 127.0.0.1 |

→ First we have to check matching bits.

Eg:



1 - care

0 - dont care

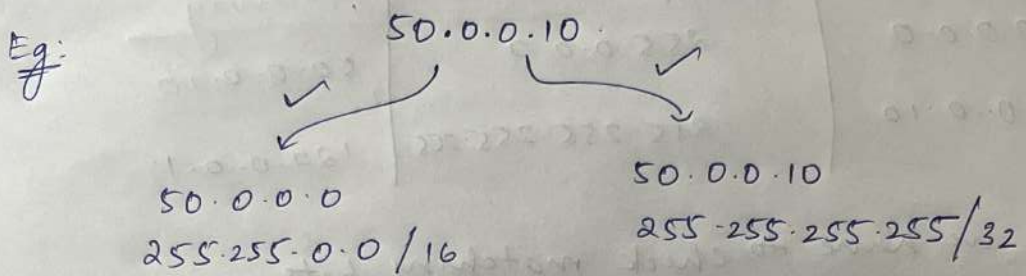
→ Here only first 2 octets are 1, so we should see whether IP address matches first two bits

→ This process happens when we ping a device. Echo is produced in app. layer → format encapsulation in pre. layer → time encap. in session layer → trans. layer → network layer - packet encapsulation. ↓

Destination IP should be checked in system route table to see through which gateway data must be sent



If the IP matches both the route, then the route with highest matching bits should be considered.



It matches both the condition, but highest matching bits route must be selected.

→ If there is a condition where we have to check our system's network connection, then we can ping using system's IP.

It is called self-communication.

→ For this purpose there is a dedicated IP for self-communication called loopback IP.

Loop-back IP: 127.0.0.1

Entire 127 series is particularly dedicated for self-communication but only 127.0.0.1 is used.